

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram and the information describing the technologies, components, data flow, and interfaces used in the proposed solution.

Example: Healthy Lifestyle and Dietary Habits Analysis using Tableau

Reference: Tableau Data Visualization and Analytics Architecture

Table-1 : Components & Technologies:

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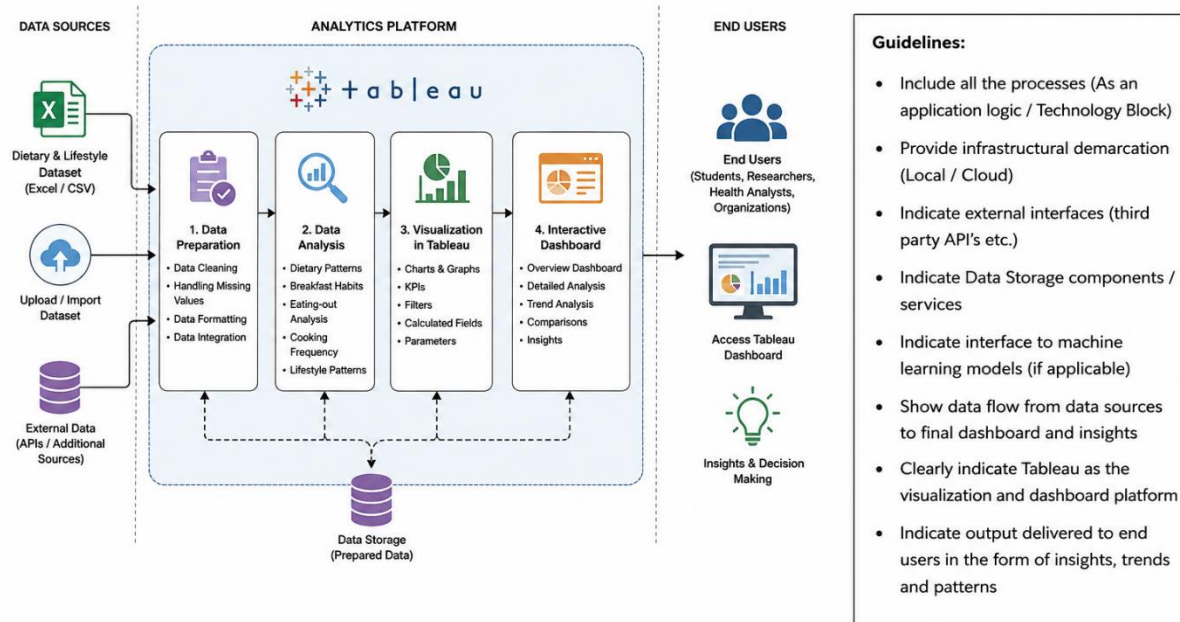


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard through which users view dietary and lifestyle analysis, charts,	Tableau Dashboard

		KPIs, filters, and insights.Web UI, Mobile App, Chatbot etc.	
2.	Application Logic-1	Data preparation and cleaning, including handling missing values, removing unnecessary data, and preparing the dataset for analysis.	Tableau / Microsoft Excel
3.	Application Logic-2	Analysis of dietary habits such as breakfast consumption, eating-out frequency, and cooking frequency.	Tableau
4.	Application Logic-3	Creation of visualizations, calculated fields, KPIs, filters, and interactive dashboard components.	Tableau
5.	Database	Stores and provides the structured dietary and lifestyle dataset used for analysis.	Not Applicable
6.	Cloud Database	Used for storing the dataset, Tableau workbook, and project-related files.	Local File System / Google Drive
7.	File Storage	No external API is required for collecting or analyzing the project data.	Not Applicable
8.	External API-1	No external API is used in the proposed solution.	Not Applicable
9.	External API-2	Machine learning is not used. The project focuses on descriptive analysis and visualization of dietary and lifestyle patterns.	Not Applicable
10.	Machine Learning Model	The project is developed and visualized using Tableau on a local system, with project files optionally stored in cloud storage.	Local System / Tableau
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

TTable-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source application framework is used. The project primarily uses Tableau for data analysis and visualization.	Tableau
2.	Security Implementations	Project files and dashboard access can be restricted to authorized users. The dataset should be handled securely and should not contain sensitive personal information.	Tableau Access Controls / File Permissions
3.	Scalable Architecture	The dashboard can be extended by adding more dietary and lifestyle records, new visualizations, filters, and KPIs as the project grows.	Tableau
4.	Availability	The dashboard can be accessed whenever the Tableau workbook and required dataset are available on the local system or shared	Tableau / Local System

S.No	Characteristics	Description	Technology
		through the appropriate Tableau environment.	
5.	Performance	The dashboard is designed using relevant visualizations, filters, and KPIs to provide analysis efficiently. The dataset can be optimized by removing unnecessary fields and records.	Tableau